

DOYUN PARK

📍 291, Daehak-ro, Yuseong-gu, Daejeon, Republic of Korea

✉ donext@kaist.ac.kr ☎ +82 10-3145-8459

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, S.Korea

Ph.D. in Graduate School of AI Semiconductor

Sep 2024 - Current

– Digital healthcare research group under the supervision of Prof. Kyeongha Kwon

M.S. in School of Electrical Engineering

Sep 2022 - Aug 2024

– Digital healthcare research group under the supervision of Prof. Kyeongha Kwon

– GPA: 4.00/4.3

B.S. in School of Electrical Engineering

Mar 2017 - Aug 2022

– GPA: 3.55/4.3 (EE GPA: 3.67/4.3)

RESEARCH INTEREST

- Wireless and Wearable Biomedical System
- Human-Computer Interaction & AI for Accessibility

PUBLICATIONS

*These authors contributed equally

1. C. Park*, **D. Y. Park***, H. Zhang*, D. S. Yang*, C. R. Redden, S. Yoo, T. Park, M. Zhou, J.-Y. Yoo, Y. Sim, A. Vázquez-Guardado, S. Y. Heo, R. Avila, Z.-B. Hou, J. Kim, H.-Y. Ahn, Y. Kim, S. Oh, J.-H. Kim, J. Baek, A. Banks, A. F. Shaaban, Y. Huang, K. Kwon, J. A. Rogers. "Adaptive electronics for photovoltaic, photoluminescent and photometric methods in power harvesting for wireless wearable sensors" *Nature Communications* (2025).
2. **D. Y. Park***, H.-M. Lee*, S.-H. Kim, Y. Sim, Y. Kang, G. Jang, S. Kim, K. Kwon, H. Ryu. "Recent Advances in Bioresorbable Biomedical Applications: From Materials to Devices" *Advanced Electronic Materials* (2025).
3. H. Ryu*, J. W. Song*, H. Luan*, Y. Sim*, S. S. Kwak, H. Jang, Y. J. Jo, H.-J. Yoon, H. Jeong, J. Shin, **D. Y. Park**, K. Kwon, G. A. Ameer, J. A. Rogers. "Materials and Device Designs for Wireless Monitoring of Temperature and Thermal Transport Properties of Wound Beds during Healing" *Advanced Healthcare Materials* (2024).
4. J. Kim*, M. I. Bury*, K. Kwon*, J.-Y. Yoo, N. V. Halstead, H.-S. Shin, S. Li, S. M. Won, M.-H. Seo, Y. Wu, **D. Y. Park**, M. Kini, J. H. Lee, S. Kim, H. Ryu, K. Yamagishi, H.-J. Yoon, S. S. Kwak, B. Kim, Y. Huang, L. C. Halliday, E. Y. Cheng, G. A. Ameer, A. K. Sharma, J. A. Rogers. "A Wireless, Implantable Bioelectronic System for Monitoring Urinary Bladder Function Following Surgical Recovery" *Proceedings of the National Academy of Sciences of the United States of America (PNAS)* (2024)

WORK IN PROGRESS

1. Y. Kang*, **D. Y. Park***, Y. Sim, G. Jang, J. Y. Kim, K. Kwon. "A wireless mask-free capnography patch for ambulatory end-tidal CO₂ and cardiopulmonary monitoring"

PROJECTS

SoundGlass: Wearable Sound-to-Color converter for hearing-impaired gamers

Jul. 2025 - Sep. 2025

– Developed a glass-compatible wearable device that 1) classifies voice/gun sounds and directions and 2) provides a real-time feedback in the form of different colors and lighting directions.

– **Won 3rd Prize at ATRAC Assistive Technology Idea Contest (2025).**

NFC-based pressure & temperature sensing platform

Aug. 2023 - Oct. 2023

– Developed a smartphone application that receives pressure/temperature data from the digital sensor through NFC.

Localization using UWB and IMU

Aug. 2023 - Dec. 2023

– Developed a distance measurement platform using an Arduino Nano 33 BLE Sense and a UWB anchor module with Kalman filter implementation.

Human motion classification application

Nov. 2022 - Nov. 2022

– Developed an Android smartphone application that classifies motion(sit/stand/jump) using deep-learning.

ChocolateIt: Screen merging application for Youtube videos

Aug. 2022 - Dec. 2022

– Developed an iOS smartphone application that merges multiple smartphone screens to form a large single screen.

Smartwatch prototype development

Mar. 2022 - Apr. 2022

– Developed a prototype smartwatch that provides a simple logo, current time, temperature, and humidity using an Arduino Nano 33 BLE Sense and an OLED module.

Remote Device Configuration using BLE

Mar. 2022 - Apr. 2022

– Developed a smartphone application and BLE SoC firmware that adjusts parameter for bladder-monitoring device.

PATENTS

K. Kwon, **D.Y. Park**, Y. Kang, "Method and apparatus for measuring biosignals." Korean Patent Application No.10-2026-0004687. Filed: Jan 9, 2026

K. Kwon, **D.Y. Park**, Y. Kang, "EtCO2 measurement system with chamberless wearable type and chamberless wearable apparatus." Korean Patent Application No.10-2024-0185252. Filed: Dec 12, 2024

K. Kwon, Y. Sim, **D.Y. Park**, C. Jung, B. Kwon, S. Lee, S.K. Kim, H. Lee, H. Rhee, S. Kim, "Electronic device and method of estimating blood flow information using the same", US Patent Application No. 18/127,445. Filed: Mar 28, 2023

SELECTED COURSEWORK

Circuits

Analog Integrated Circuits Digital Integrated Circuits Low Noise Electronic Circuits
Advanced Electronic Circuits Digital Electronic Circuits Digital System Design Analog System Design

Electrical Engineering & Computer Science

Embedded Systems Data Structures and Algorithms for EE Data Structure
Special Topics in ECE (Mobile Computing, Sensing, Learning, and Interactions)
Special Topics in ECE (Mobile Computing and Systems for Intelligent Living)
Special Topics in EE (Machine Learning Basics and Practices)

AWARDS AND HONORS

3rd Prize, Assistive Technology Idea Contest (Designated Topic Track)

Nov. 2025

ATRAC (Assistive Technology Research and Application Center), Gyeonggi Province

2nd Prize, 2026 KAIST OpenSource Hackathon 301

Jun. 2026

SW Education Center, KAIST

SKILLS

C, C++, Java, MATLAB, Python, Flutter

TEACHING EXPERIENCES

Teaching Assistant School of Electrical Engineering, KAIST

Responsibilities include grading assignments, conducting recitation sessions, and providing student support

- Digital System Design (Undergraduate) Spring 2023 - Spring 2024, Spring 2025

- Future Semiconductor Circuit Technology (Graduate) Fall 2024

- Special Topics in ECE 'Mobile Computing, Sensing, Learning, and Interactions' (Graduate) Fall 2025